

Integrating Ethics in Public Health Guideline Development: Case Study of Guidelines on Respiratory Isolation for Persons With Tuberculosis in Community Settings

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Public health interventions often involve explicit trade-offs in which the health of the many must be weighed against burdens imposed on individuals. We describe development of public health guidelines for respiratory isolation in community settings for persons with tuberculosis. While stopping the spread of disease is a core moral imperative in public health, the duty to prevent disease transmission does not supersede all other considerations. Community well-being must be balanced with individual well-being, liberty, and social justice. In response to these challenges, the National Tuberculosis Coalition of America's 2024 guidelines for persons with tuberculosis in community settings were developed using a modified GRADE approach supported by a complementary, comprehensive, and context-specific ethical framework. By addressing the distinct roles that evidence (subject to uncertainty), values, justificatory conditions, and procedural legitimacy all play in ethical guideline development, we promote rigor and transparency in the integration of ethics in public health guidelines.

Keywords. tuberculosis; public health; isolation; ethics; policy.

Public health policies are often articulated through laws, regulations, and guidelines developed by governmental and nongovernmental organizations then implemented by public health practitioners, often within local and state public health departments. The process of public health policy formulation and implementation is complex and often lacking in transparency.

These challenges were evident during the coronavirus disease 2019 (COVID-19) pandemic, where the rationale for public health recommendations was not always articulated; attempts to follow the evidence did not always acknowledge scientific uncertainty; and some priorities (including impacts on the economies, education, or mental health) were not always acknowledged in decision-making [1]. In the wake of these challenges, trust in public health guidelines has been strained.

We present a case study of public health guideline development about respiratory isolation and restrictions (RIR) in community settings for persons with tuberculosis in the United States by the National Tuberculosis Coalition of America

(NTCA) [2]. RIR has been used for decades, but has never been formally recommended in national-level guidance from the Centers for Disease Control and Prevention (CDC), leading to a patchwork of regional policies with heterogeneous implementation [3, 4]. The NTCA sought to develop evidence-based and ethics-informed guidelines to support public health officers within their communities. We provide an overview of how public health ethics can be integrated into a guideline development with the 2024 NTCA Guidelines on Respiratory Isolation and Restrictions for Persons With Tuberculosis in Community Settings (NTCA Guidelines) as a model.

THE FOUNDATIONAL ROLE OF ETHICS IN GUIDELINE DEVELOPMENT

A guideline is intended to direct users toward the “right” action (whether a clinical decision, test, treatment, or public health intervention), but “right” is a complex concept. Policymakers must synthesize data to understand what is “right” in the sense that it is scientifically correct or true, often through data review [5]. Scientific data have limitations, and policymakers must navigate uncertainty about interventions and their predicted effects. Policymakers must also decide which effects should be promoted because they are good or beneficial, and which should be avoided or minimized as negative or harmful. These judgements require consideration of ethical values. Interventions with both positive and negative effects require that policymakers weigh whether the “good” effects justify

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the “bad.” How policymakers make judgments is also subject to ethical requirements for proper conduct. This organizational framework (Table 1) highlights distinct roles that evidence (and uncertainty), values, justificatory conditions, and procedural legitimacy play in ethical guideline development.

Clinical practice guidelines (CPG) increasingly use the GRADE (grading of recommendations, assessment, development, and evaluations) process, which promotes formal assessment of the certainty of evidence behind recommendations [6]. Scientific data are used to evaluate interventions’ effects in terms of health outcomes, toxicities, and costs. Effects are classified as “benefits” or “harms” based on whether they promote or detract from patients’ well-being. Within this framework, “strong recommendations” are those where benefits significantly outweigh harms.

CPGs rarely deal with complex trade-offs between individual and community interests required for public health policies. Consequently, the role of ethical values in clinical recommendations is often obscured because they focus on one value—patients’ well-being—and scientific data can be interpreted as directly describing benefits and harms to the patient.

Public health guidelines confront trade-offs between individual and community interests explicitly. Public health ethics emerged in the 20th century, building on foundations established in bioethics, clinical ethics, and research ethics [7]. Development of public health interventions for infectious diseases poses several key challenges: First, the US public health system is a patchwork of local and state departments and federal health agencies, each empowered to make public health policies [4]. Second, there is not a standard methodology by which public health guidelines are developed. The GRADE process that is often productive for CPGs is less suited to decisions about public health interventions [8, 9]. Even with adaptations of the GRADE process to public health, addressing questions about *transmissible* diseases remains a challenge because of substantial trade-offs between individual and community interests or other values, combined with uncertainty. Third, there is a lack of training or pragmatic guidance for public health practitioners on how to incorporate ethics into public health decision-making.

CASE STUDY OF NTCA GUIDELINES FOR USE OF RESPIRATORY ISOLATION AND RESTRICTIONS FOR PERSONS WITH TUBERCULOSIS IN COMMUNITY SETTINGS

A NTCA Guideline Development Group (GDG) was convened to develop recommendations for community-based RIR for persons with tuberculosis in the United States, where no prior national-level guidelines existed. The NTCA was a unique entity for this task, representing the largest professional organization of public health nurses, clinicians, epidemiologists, tuberculosis survivors, and public health professionals in local and state tuberculosis programs across the United States. An independent Evidence Synthesis Group synthesized literature regarding tuberculosis transmission and patient and population outcomes related to community-based RIR.

A priori, there was recognition that moving from scientific evidence to policy would entail ethically complex judgments, so the NTCA included bioethicists to help the GDG construct a context-specific ethical framework to support decision-making. This framework is influenced by precedent frameworks by Kass [10], Childress [11], Thompson [12], Baum [13], and the Nuffield Council on Bioethics [14], while also capturing topic-specific expertise and experience of GDG members and establishing a shared ethics vocabulary. The creation of a context-specific ethical framework concurrently with guideline development represents a novel application of these resources that could be utilized by future guideline-development groups dealing with ethically complex recommendations.

Setting the Stage

At the onset of guideline development, the GDG collaboratively created a statement of purpose:

“A community-based tuberculosis isolation policy is needed because tuberculosis is an airborne infection capable of spreading through communities, and an isolation policy has the potential to reduce or stop the spread of tuberculosis. However, current practices may prioritize public

Table 1. Dimensions of “Right” That Come Together to Support Recommendations in Guidelines

Dimension	What it Means	What to Ask	Concept to Apply
Right (correct)	When action X is taken, we can predict that outcome Y and outcome Z are likely to happen	What do scientific data tell us about the likely effects of RIR?	Approaches to uncertainty (evidence synthesis)
Right (good)	Outcome Y is a positively valued effect that we seek to bring about; outcome Z is a negatively valued effect that we seek to avoid	Which effects of RIR are good and align with our values, and which are bad or detrimental to our values?	Ethical values
Right (fair)	When action X is taken, the negatively valued effect Z is sufficiently balanced by the positively valued effect Y	Do the positive effects of RIR justify the negative effects of RIR?	Justificatory conditions
Right (proper)	In developing the recommendation for action X, we have acted appropriately and shown good conduct	Is the policymaking process happening according to appropriate standards?	Standards for procedural legitimacy

Abbreviation: RIR, respiratory isolation and restrictions.

Table 2. An Ethical Framework to Support Public Health Guideline Development

Domain	Term	Explanation
Approaches to uncertainty Support policymakers in interpreting limited scientific data	Abstention	Make no recommendation, but be mindful of what will happen in the absence of guidance
	Prioritization	Default to prioritizing a single value or good
	Evidence grading	Transparently describe the limitations of scientific data
	Practical wisdom	Critically examine and interpret limited data using expertise gained through relevant training and personal experience
Ethical values Support policymakers in deciding which effects of a policy decision would be desirable and should be promoted, or undesirable and should be avoided/minimized	Well-being	Individual
		Health: individual health and safety outcomes
		Livelihood: access to work, school, housing, food, and basic resources
		Social relations: access to stable and supportive social network
		Esteem: self-esteem and dignity
	Community	Public health: community and population health and safety outcomes
		Economic productivity: community-level economic stability; GDP
		Social cohesion: strength and stability of shared community bonds
		Community identity: sense of national, regional, or group identity and esteem
	Justice	Recognition: be aware of differences in identity and experience
		Distributive: seek a fair distribution of benefits and burdens
		Procedural: ensure due process
	Liberty	Freedom from obstructive interference, without exclusion from beneficial or empowering resources
		Incorporate opportunities for choice
		Offer compensatory resources
Justificatory conditions Support policymakers in determining whether a compromise between values is acceptable and fair	Necessity	Infringement on one value to promote another must be necessary or unavoidable
	Proportionality	Infringement on one value must be balanced by at least qualitatively proportional promotion of another
	Least infringement	Refine the conditions of a proposed compromise to minimize infringement on any value
	Lines in the sand	Identify the boundaries of a pluralistic framework by defining compromises that would be unacceptable because they violate fundamental rights or obligations
Standards for procedural legitimacy Set expectations for how policies should be made and how policymakers should act	Representation	Include representatives with diverse backgrounds and experiences related to the policy topic and its expected impact
	Transparency	Proactively promote openness to examination and critique by experts, colleagues, stakeholders, and the public
	Management of conflicts and biases	Report potential conflicts of interest and mitigate conflicts with practices such as disclosure, recusal, or divestment
	Humility	Be aware of the limitations of policymakers relative to the ambitious task of policymaking; use self-reflection to observe and moderate one's individual influence in policymaking
	Clarity of purpose	Begin with a clear sense of what the policy is intended to achieve or bring about in the world
		Critically examine the finished policy for alignment with its stated purpose

health and safety to an extent that is detrimental to individual well-being, liberty and social justice; current practices may be inconsistent and varied, which is detrimental to social justice and procedural legitimacy; and current practices may not reflect all available scientific data.”

Several ethical concepts are implicit (or explicit) in this statement of purpose. First, stopping the spread of disease (preventing harms) is a core moral imperative in public health practice. Second, duty to prevent infectious disease transmission does not supersede all other considerations. Instead, a pluralistic system of values including the value of public health with coequal values of community and individual well-being, liberty, and justice should be considered and balanced. A challenge forecasted in this value pluralism is the need to balance promotion of one value with detriment to others. A further challenge is the need for (but limitations of) scientific data.

Construction of an Ethical Framework for the NTCA Guidelines

The GDG next created an ethical framework consistent with this statement and responsive to its challenges. Ethical values, justificatory conditions, approaches to uncertainty, and standards for procedural legitimacy were proposed and refined through iterative discussion and revision (Table 2). These procedures resemble the iterative, discursive generation of a bioethics framework for pandemic influenza policy described by Thompson [12].

Standards for Procedural Legitimacy

Standards for procedural legitimacy address the question of how policymakers should approach their task. These standards call upon the GDG to examine their own conduct and describe benchmarks to which the GDG can expect to be held by external observers [11].

Representation. This standard calls for collaboration among people with diverse experiences related to tuberculosis isolation

policy. Meaningful representation requires members be equal collaborators, not just witnesses, sources of information, or endorsers of final decisions.

The abstract populations subject to public health actions is often a barrier to representation [7]. The NTCA GDG included tuberculosis survivors with experiences of isolation but did not directly include community members at risk for exposure to tuberculosis. However, the GDG included tuberculosis clinicians, epidemiologists, and health officers with insights into public health responsibilities, and solicited input through electronic surveys from public health practitioners at the NTCA 2023 Conference.

Transparency. This standard calls for openness to critique of the policymaking process and the finished policy by experts, stakeholders, and the public. Transparency is achieved by disseminating the policy and policymaking process (including this framework) in diverse and accessible formats.

During guideline development from 2022 to 2024, the evidence synthesis and ethical framework were presented at a national conference; and draft guidelines were shared at regional tuberculosis health care worker trainings, CDC staff educational conferences, and an NTCA-sponsored Community of Practice webinar to promote transparency, incorporate feedback, and increase confidence in the process by which recommendations were developed.

Like representation, transparency is a mechanism for stakeholder engagement in policymaking. Precedent frameworks propose related concepts: accountability [13], responsiveness [12], and public justification [11].

Management of Biases and Conflicts of Interest. This standard calls for acknowledgement of potential conflicts and management through actions such as disclosure, recusal, or divestment, even when few or no conflicts are anticipated.

Humility. The standard of humility calls for awareness of one's own limitations as a policymaker relative to the ambitious task of guideline development. Humility as a self-reflective virtue is not seen in other public health ethics frameworks, which tend to emphasize checks on authoritarianism rather than checks on hubris [7]. Its inclusion here stems from recognition that the GDG was producing both this ethical framework and the public health guideline for which the framework was developed. In setting standards for procedural legitimacy as a part of this framework, the GDG set standards for its own conduct, inviting self-reflection.

Clarity of Purpose and Alignment With Purpose. This standard is evident in many precedent frameworks and is partly fulfilled in our statement of purpose. A finished policy should be clearly aligned with its stated purpose.

Table 3. Dimensions of Well-being Considered by the National Tuberculosis Coalition of America Guideline Development Group

Dimension	Individual	Community/Public
Health	Personal health outcomes related to tuberculosis or other conditions	Public health outcomes including tuberculosis transmission and active tuberculosis incidence
Livelihood	Individual or household financial stability, employment, accesses to resources such as food and housing	Economic outcomes at local, regional, national, or global levels
Social relations	Relationships with family, friends, and colleagues Access to a social support network	Social cohesion at the local, regional, national, or global levels
Esteem	Self-esteem, feelings of pride and dignity	Community or group identity including national identity

Ethical Values

Ethical values describe goods that we seek to promote (or not erode) and have their roots in principle-based bioethics [7].

Well-being. Well-being is a multidimensional concept encompassing individual and public interests in terms of health, livelihood, social relations, and esteem (Table 3). These domains are interconnected—self-esteem shapes social relations which shape livelihood which shapes health; social cohesion shapes community and individual health, etc. By valuing individual well-being in parallel with community well-being, we disrupt a historical tendency to reduce public health ethics to a conflict between public health and liberty.

In addition to the health of the community, economic productivity and social cohesion are valued aspects of public well-being that can be affected by public health policies. Social cohesion is a long-recognized health determinant [15, 10], and has resurfaced as a valued good affected by health policy in the COVID-19 pandemic [16]. The NTCA GDG considered these dimensions of both individual and community well-being in its guideline development process.

Justice. Recognition, distributive, and procedural justice are each important for fully integrating the value of justice in public health policy development. Recognition justice calls for awareness of and responsiveness to differences in identity and experience [17]. For example, NTCA Guidelines recommend additional consideration in cases where there is risk of tuberculosis transmission to vulnerable populations like immunosuppressed individuals and children, recognizing these groups as vulnerable and adapting recommendations to ensure these unique needs are not disregarded. Distributive justice calls for fair distribution of expected benefits and burdens, generally in terms of resources [10, 11]. Recognition and distributive justice have parallels in our value of well-being; recognition justice deals more closely with dignity and social

relations; distributive justice deals more closely with health and livelihood.

Inequities in these domains already exist in society. Public health policies that exacerbate existing inequities are detrimental to justice and require explanation or justification. Societal inequities factor into who contracts tuberculosis, and RIR can exacerbate inequities through negative impacts on livelihood, social relations, and esteem. Negative effects are anticipated to be more pronounced among already marginalized or vulnerable populations because of barriers to secure isolation, loss of employment, effects on education, and stigmatization. At the same time, *effective* RIR can reduce tuberculosis transmission within networks of individuals at higher risk, partially mitigating inequities through positive effects on health [18]. The GDG expressed concern that the net impact of RIR may be detrimental to justice, necessitating explanation and mitigation, an application of the justificatory condition of least infringement.

Finally, procedural justice calls for due process when RIR are being imposed [11]. For example, NTCA Guidelines integrate process considerations in an implementation aid recommending additional review and expert consultation in cases where RIR extends beyond 14 days.

Liberty. The value of liberty calls for freedom to order one's life as one sees fit, without infringing upon the freedom of others. The GDG considered liberty to involve freedom from *obstructive* interference, but not exclusion from beneficial or empowering resources [12]. Although obstructive interference is inherent in RIR, liberty is partly preserved by allowing opportunities for choice (ie, allowing outdoor activities while restricting indoor or higher-risk activities), and offering compensatory resources (ie, food, housing, remote access to school and employment) [19].

Justificatory Conditions

In a pluralistic system with potentially competing values, the GDG identified conditions that would justify infringement upon one value to promote another, adapted from Childress [11].

Necessity. To be justified, a compromise to one value with the goal of promoting another value must be *necessary* to achieve that goal.

Proportionality. A compromise to one value for the sake of another must be at least a balanced compromise. The GDG carefully considered the relative priority of positive effects of RIR like reducing tuberculosis transmission and negative effects such as harm to individual mental health, exacerbation of inequities, and restrictions on liberty. There was concern that recommending protracted RIR might achieve limited or no benefit to community well-being at *disproportionate* cost to individual well-being, justice, and liberty. Alternatively,

recommending no RIR would avoid direct infringement upon individual well-being, justice, and liberty, but at the cost of disproportionate detriment to community well-being. The GDG sought to determine conditions for recommending RIR that would promote public health while ensuring no extreme or disproportionate infringements upon other values.

Least Infringement. When a policy infringes on any value, steps should still be taken to minimize the infringement. The condition of least infringement can be applied to refine details of proposed compromises that meet conditions of necessity and proportionality. The GDG sought to define conditions of RIR that would minimize infringements on individual well-being, justice, and liberty.

Lines in the Sand. Although this framework comprises coequal values with no single overriding imperative, there may be boundaries to such a framework, understood as "lines in the sand." Lines in the sand would be infringements that could not be justified, even if they appear to meet the standards of necessity, proportionality, and least infringement, and even if they would substantially promote one or more of our values. In a pluralistic framework, lines in the sand should be uncommon and extreme. In the context of the NTCA Guidelines, the GDG determined that individuals have a fundamental right to access health care and emergency services, and therefore no one should be restricted from accessing these services.

Approaches to Uncertainty

Ethical values, justificatory conditions, and standards for procedural legitimacy establish a framework for determining the *ethical* (good, fair, proper) dimensions of guideline development. However, guideline development is also subject to ethically important challenges with determining the *factual* (scientific) basis for recommendations, because of uncertainty [20].

Scientific uncertainty has a cascading effect. It challenges our ability to precisely forecast the effects of a recommendation, which affects our ability to predict whether the recommendation will promote or be detrimental to our ethical values and to what extent, which in turn affects our ability to decide if justificatory conditions like necessity and proportionality are met.

In developing the NTCA Guidelines, uncertainty was evident in several areas. The evidence synthesis team judged there to be "very low" certainty of evidence regarding the effects of community-based isolation for preventing tuberculosis transmission, incidence, or mortality. Despite "moderate" certainty of evidence that treatment reduces infectiousness, there was uncertainty about the specific duration of treatment at which most people will become noninfectious [2]. In response to these challenges, the GDG set out to define approaches to uncertainty and unmask the reasoning linking limited scientific data to recommendations.

Abstention. Faced with uncertainty, one option is to give no recommendation. However, the GDG was mindful of what may happen in the absence of recommendations: that is, perpetuation of a status quo, extrapolation from related but outdated recommendations, and continuation of inconsistent practices.

Prioritization. Policymakers facing uncertainty may choose to prioritize one value or end point over others. For example, a common default in public health is to prioritize community well-being (to err on the side of “over-isolation”), an extreme precautionary approach [14]. Such an approach would run counter to a framework of *coequal* values, in which the relative priority of effects like promoting community well-being or detracting from individual well-being is decided not by default but by justificatory conditions.

Grading Certainty of Evidence. Evidence grading promotes transparency by acknowledging the limitations of evidence underlying recommendations. The NTCA Guidelines use the GRADE approach for assessing certainty of evidence, accounting for risk of bias, indirectness, and inconsistency [6]. Evidence grading *describes* uncertainty, but it does not necessarily dictate how to *decide* under conditions of uncertainty.

The GDG also used evidence to decision (ETD) tables according to the GRADE approach adapted for public health questions. The ETD process incorporates evidence grading and additional

domains including acceptability, impact on equity, and feasibility [21]. Evidence grading alone does not determine the final recommendation (recommendations are not constrained to always align with the highest-certainty evidence). Instead, grading is integrated with policymakers’ judgements about ethical values (desirable or undesirable effects), justificatory conditions (balance among effects), and contextual factors. This ethical framework supported the GDG in such ethically complex judgments and is complementary to the GRADE ETD process.

Practical Wisdom. The GDG leaned on the practical wisdom of representatives with relevant training and experience to interpret limited evidence in the context of uncertainty. A seasoned practitioner’s expert opinion is a kind of practical wisdom; so is an individual’s knowledge about the burdens of isolation after experiencing isolation firsthand. Practical wisdom involves critical interpretation of scientific data to navigate uncertainty but does not supersede scientific data when data are available, and certainty is high.

USING THE FRAMEWORK TO DEVELOP EVIDENCE-INFORMED AND ETHICS-INFORMED GUIDELINES

RIR can promote public health by reducing tuberculosis transmission (and potential health consequences of tuberculosis

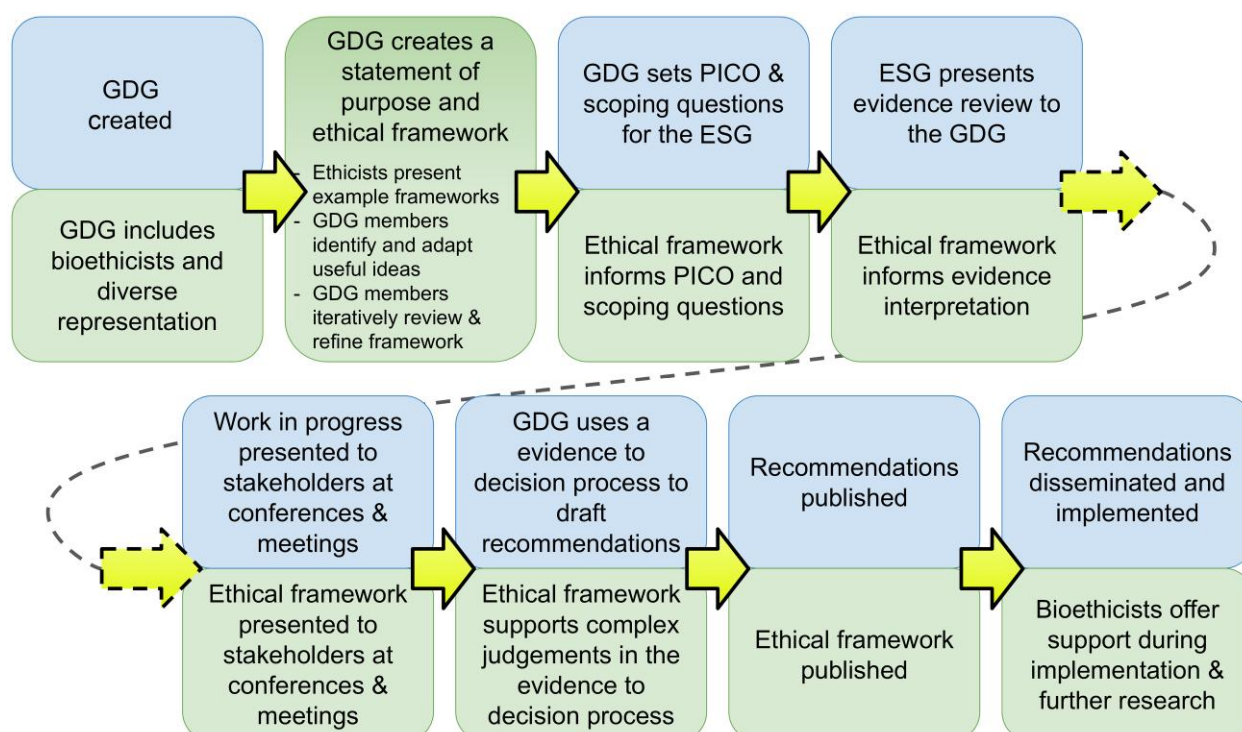


Figure 1. Roadmap for integrating bioethics in the guideline development process. Abbreviations: ESG, evidence synthesis group; GDG, guideline development group; PICO, population, intervention, comparison, and outcomes.

infection), although there is low certainty of evidence about these effects. On the other hand, RIR are detrimental to individual well-being in terms of livelihood, social relations, and esteem; to justice; and to liberty; with moderate certainty of evidence about negative effects on mental health and stigma. Compromising the values of individual well-being, justice, and liberty can be justifiable if doing so is necessary to achieve an at least proportional benefit to public health. If imposed, the conditions of RIR should be defined such that they cause the least infringement possible on individual well-being, justice, and liberty.

There is uncertainty as to the precise moment when a person with tuberculosis becomes unable to infect others, but moderate certainty of scientific evidence supports that infectiousness declines rapidly and steadily over time with effective treatment (and that sputum smear microscopy does not predict infectiousness during treatment). By contrast, evidence and practical wisdom support that burdens of RIR increase over time. At some point along these trajectories, the burdens of RIR on individuals will not be justified by at least proportional benefits to public health. An ethical isolation policy, according to this framework, would not impose RIR beyond that point.

The NTCA Guidelines recommend ending restrictions in most cases after 5 days of effective treatment, irrespective of sputum smear microscopy, with individualized exceptions. This differs from historic practices in US health departments, and recommendations in some countries where thresholds for deisolation involve a “minimum of 2 weeks of effective therapy... and 3 consecutive negative acid-fast bacilli sputum smears.” [2]

This ethical framework developed and used by the NTCA GDG supports and explains the difference. First, we do not treat public health as an overriding good, inherently more valuable than other goods of individual well-being, justice, and liberty, but consider these coequal values needing balance. Second, we do not default to prioritizing public health under conditions of scientific uncertainty about infectiousness, using an extreme precautionary approach. Rather, the GDG integrated available scientific evidence, practical wisdom, ethical values, and justificatory conditions to formulate more nuanced judgments.

CONCLUSION

We describe a multidisciplinary collaboration to concurrently develop both a context-specific ethical framework and a public health guideline according to that framework (Figure 1). We undertook these tasks in parallel not to serve as arbiters of our own ethical integrity but to foster critical ethical reasoning for policymakers, integrate ethical considerations in all phases of guideline development, synchronize language for discussing ethical issues despite diverse backgrounds, and ensure our

ethical framework’s responsiveness to challenges arising during policymaking. The overarching structure of our framework complements existing guideline development processes, providing a roadmap to address challenges from scientific uncertainty, desirable and undesirable effects experienced by different target populations (eg, communities exposed to infectious diseases and individuals having an infectious disease), balancing multiple effects, and promoting good conduct in policymaking. Future groups confronted with ethically complex public health questions could develop their own context-specific ethical frameworks according to this example by delineating approaches to uncertainty, ethical values, justificatory conditions, and standards for procedural legitimacy through facilitated discussions among group members.

Notes

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